

OPF Problem Variants and References

Overview

This document catalogs the structural variants of Optimal Power Flow (OPF) problems identified in our work **ProOPF: Benchmark and Advance LLM Competence in Domain-Specialized Optimal Power Flow Modeling for Power Systems**. The variants are organized into three categories: Decision Variable Extensions, Objective Function Extensions, and Constraint Extensions. Each table includes a references column for tracking the key papers that introduce or study these variants.

1 Decision Variable Extensions

1.1 Part 1

Table 1: Decision Variable Variants in OPF Problems (Part 1) with References

ID	Problem Type	Formulation	Descriptions	References
1	AC OPF	$\min_{P_g, Q_g, V, \theta} \sum_{g \in \mathcal{G}} C_g(P_g)$ subject to: $P_g - P_d = \sum_k V_i V_k Y_{ik} \cos(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $Q_g - Q_d = \sum_k V_i V_k Y_{ik} \sin(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $P_g^{\min} \leq P_g \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_g \leq Q_g^{\max}, \quad \forall g$ $V^{\min} \leq V \leq V^{\max}, \quad S_{ik} \leq S_{ik}^{\max}, \quad \forall i, k$	Canonical OPF formulation studied in this work; optimizes generator dispatch (P_g, Q_g) and bus voltages (V, θ) subject to nonlinear AC power flow equations and operational limits; serves as the base problem formulation upon which objective function and constraint extensions are applied	[1], [2], [3], [4], [5], [6], [7], [8], [9]
2	DC OPF	$\min_{P_g, \theta} \sum_{g \in \mathcal{G}} C_g(P_g)$ subject to: $\sum_{g \in \mathcal{G}_i} P_g - P_{d,i} = \sum_{k \in \mathcal{N}_i} B_{ik}(\theta_i - \theta_k), \quad \forall i \in \mathcal{N}$ $ B_{ik}(\theta_i - \theta_k) \leq F_{ik}^{\max}, \quad \forall (i, k) \in \mathcal{L}$ $P_g^{\min} \leq P_g \leq P_g^{\max}, \quad \forall g$ $\theta_{\text{ref}} = 0$	Simplifies AC-OPF by eliminating voltage magnitude and reactive power variables; linearizes power flow using small-angle approximation; widely used in large-scale system operations, market clearing, and real-time dispatch due to computational efficiency	[10], [11], [12], [13], [14], [15], [16]
3	Multi-period OPF	$\min_{P_g, Q_g, V, \theta} \sum_{t=1}^T \sum_{g \in \mathcal{G}} C_g(P_{gt})$ subject to: $P_{gt} - P_{dt} = \sum_k V_{it} V_{kt} Y_{ik} \cos(\theta_{it} - \theta_{kt} - \alpha_{ik}), \quad \forall i, t$ $Q_{gt} - Q_{dt} = \sum_k V_{it} V_{kt} Y_{ik} \sin(\theta_{it} - \theta_{kt} - \alpha_{ik}), \quad \forall i, t$ $P_g^{\min} \leq P_{gt} \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_{gt} \leq Q_g^{\max}, \quad \forall g, t$ $V^{\min} \leq V_{it} \leq V^{\max}, \quad S_{ikt} \leq S_{ik}^{\max}, \quad \forall i, k, t$ $P_{gt} - P_{g,t-1} \leq R_g^{\text{up}}, \quad P_{g,t-1} - P_{gt} \leq R_g^{\text{dn}}, \quad \forall g, t \geq 2$	Extends single-period OPF over planning horizon T by coupling consecutive periods via generator ramping constraints; optimizes generation schedule considering temporal variations in load and renewable generation; units remain committed throughout horizon	[17], [18], [19], [20], [21]

1.2 Part 2

Table 2: Decision Variable Variants in OPF Problems (Part 2) with References

ID	Problem Type	Formulation	Descriptions	References
4	Multi-period unit commitment	$\min_{u, P_g, Q_g, V, \theta} \sum_{t=1}^T \sum_{g \in \mathcal{G}} [C_g(P_{gt})u_{gt} + SC_g(u_{gt}, u_{g,t-1})]$ where $SC_g = S_g^{\text{up}}(1 - u_{g,t-1})u_{gt} + S_g^{\text{dn}}u_{g,t-1}(1 - u_{gt})$ subject to: AC power flow equations $\forall t$ $u_{gt}P_g^{\min} \leq P_{gt} \leq u_{gt}P_g^{\max}, \quad u_{gt}Q_g^{\min} \leq Q_{gt} \leq u_{gt}Q_g^{\max}, \quad \forall g, t$ $V^{\min} \leq V_{it} \leq V^{\max}, \quad S_{ikt} \leq S_{ik}^{\max}, \quad \forall i, k, t$ $P_{gt} - P_{g,t-1} \leq R_g^{\text{up}}u_{gt}, \quad P_{g,t-1} - P_{gt} \leq R_g^{\text{dn}}u_{g,t-1}, \quad \forall g, t$ $\sum_{\tau=t-T_g^{\text{up}}+1}^t (1 - u_{g\tau}) = 0, \quad \sum_{\tau=t-T_g^{\text{dn}}+1}^t u_{g\tau} = 0, \quad \forall g, t$ $u_{gt} \in \{0, 1\}, \quad \forall g, t$	Integrates unit commitment with multi-period dispatch optimization; determines optimal on/off schedule and power output for each generator over time horizon; enforces inter-temporal constraints including minimum up/down times and startup-dependent ramping limits	[26], [27], [28]
5	Economic dispatch	$\min_{P_g} \sum_{g \in \mathcal{G}} C_g(P_g)$ subject to: $\sum_{g \in \mathcal{G}} P_g = \sum_{d \in \mathcal{D}} P_d$ $P_g^{\min} \leq P_g \leq P_g^{\max}, \quad \forall g$	Simplest formulation that determines generator active power outputs to satisfy total demand at minimum cost; neglects network topology, power flow constraints, and reactive power; provides fast dispatch decisions assuming transmission capacity is sufficient; serves as initial solution for more detailed OPF models	[29], [30], [31], [32], [29], [33], [34]

1.3 Part 3

Table 3: Decision Variable Variants in OPF Problems (Part 3) with References

ID	Problem Type	Formulation	Descriptions	References
6	Optimal transmission switching	$\min_{P_g, Q_g, V, \theta, z} \sum_{g \in \mathcal{G}} C_g(P_g)$ subject to: $P_g - P_d = \sum_k V_i V_k Y_{ik} z_{ik} \cos(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $Q_g - Q_d = \sum_k V_i V_k Y_{ik} z_{ik} \sin(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $ I_{ik} \leq z_{ik} I_{ik}^{\max}, \quad \forall (i, k) \in \mathcal{L}$ $P_g^{\min} \leq P_g \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_g \leq Q_g^{\max}, \quad V^{\min} \leq V \leq V^{\max}$ $z_{ik} \in \{0, 1\}, \quad \forall (i, k) \in \mathcal{L}$	Introduces binary variables for line switching status to enable network topology control; allows strategic line outages to relieve congestion and reduce generation cost; results in a computationally challenging mixed-integer nonlinear program (MINLP)	[13], [35], [36]
7	Load shedding	$\min_{P_g, Q_g, V, \theta, L^{\text{shed}}} \sum_{g \in \mathcal{G}} C_g(P_g) + \sum_{d \in \mathcal{D}} W_d L_d^{\text{shed}}$ subject to: $P_g - P_d + L_d^{\text{shed}} = \sum_k V_i V_k Y_{ik} \cos(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $Q_g - Q_d = \sum_k V_i V_k Y_{ik} \sin(\theta_i - \theta_k - \alpha_{ik}), \quad \forall i$ $P_g^{\min} \leq P_g \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_g \leq Q_g^{\max}, \quad V^{\min} \leq V \leq V^{\max}$ $0 \leq L_d^{\text{shed}} \leq L_d, \quad \forall d$	Introduces continuous load shedding variables L_d^{shed} to ensure problem feasibility when generation capacity is insufficient, or network constraints are too restrictive; assigns high penalty cost W_d to represent the severe economic and social impact of unserved energy	[4], [37], [38], [39]

2 Objective Function Extensions

2.1 Part 1

Table 4: Objective Function Variants in OPF Problems (Part 1) with References

ID	Objective Name	Formulation	Descriptions	References
1	Real power redispatch cost (RPR)	$RPR = \sum_{i \in N_G} (c_{it}^u P_{G,it}^u + c_{it}^d P_{G,it}^d)$	Minimize magnitude and cost of real-power adjustments between reference schedule and redispatched solution	[40], [41], [42], [12], [43], [44]
2	Reactive power generation cost (RGC)	$RGC = \sum_{i \in N_G} (a_Q Q_{G,i}^2 + b_Q Q_{G,i} + c_Q)$	Minimize total cost of providing reactive power support from generators	[45], [46]
3	Real power procurement cost (RPC)	$RPC = \sum_{i=1}^{N_{G_p}} P_{pi} \lambda$	Minimize total payment for purchasing active power from external sources at price λ	[47], [48], [49], [50]
4	Reactive power procurement cost (QPC)	$QPC = \sum_{n \in N_B} \sum_{m \in \Omega_n} V_n^2 \frac{Y_{Ch,nm}}{2}$	Minimize reactive power procurement from shunt capacitors to support bus voltage magnitudes	[51], [52], [53], [54], [55]
5	Real power reserve cost (RRC)	$RRC = \sum_{i \in I_{SP}} f_{s,i}(R_{s,i}) + \sum_{i \in I_{RU}} f_{u,i}(R_{u,i})$	Minimize capacity payment for multiple types of active power reserves (up, down, spinning, non-spinning)	[56], [57], [58]
6	Reactive reserve insufficiency cost (RRI)	$RRI = - \sum_{i \in N_G} w_i (Q_{G,i}^{\max} - Q_{G,i})$	Minimize negative reactive power reserve margin (equivalently maximize reserve) to enhance voltage security	[59], [60], [61]
7	System congestion cost (SCC)	$SCC = \frac{1}{N_T} \sum_{l=1}^{N_T} \frac{S_l}{S_l^{\max}}$	Minimize average utilization level of transmission elements to enhance loadability and reduce congestion	[2], [62], [63]
8	Voltage stability margin cost (VSC)	$VSC = \max(L_i), \quad L_j = \left 1 - \sum F_{ji} \frac{V_i}{V_j} \right $	Minimize maximum L-index to increase distance to voltage-collapse points and improve stability margin	[64], [51], [65]

2.2 Part 2

Table 5: Objective Function Variants in OPF Problems (Part 2) with References

ID	Objective Name	Formulation	Descriptions	References
9	Active power loss cost (PLC)	$PLC = \sum_{i=1}^{N_G} P_{Gi} - \sum_{j=1}^{N_L} P_{Lj}$	Minimize total active power losses in transmission network to improve system efficiency	[35], [66], [67]
10	Environmental emission cost (EEC)	$EEC = \sum_{i,t} (\alpha_i P_{G,it}^2 + \beta_i P_{G,it} + \gamma_i + \zeta_i e^{\lambda_i P_{G,it}})$	Minimize total pollutant emissions (NO _x , SO ₂ , CO ₂) using quadratic-plus-exponential emission functions	[68], [64]
11	Negative transfer capability cost (NTC)	$NTC = -\lambda \sum_{i \in S_L} b_{Pi}$	Minimize negative transfer capability (equivalently maximize load transfer capacity) to estimate available transfer capability	[69], [70], [71], [72], [73], [74]
12	Negative social welfare cost (NSW)	$NSW = \sum_{j \in G} C_j(x_j) - \sum_{i \in C} B_i(x_i)$	Minimize generation cost minus consumer utility (equivalently maximize social welfare) to achieve economic efficiency	[14], [75], [76], [77], [11]
13	Voltage deviation penalty cost (VDP)	$VDP = C_{\text{gen}}(P_G) + \beta \sum_{i=1}^N (V_i - 1.0)^2$	Minimize generation cost plus penalty on squared voltage deviations from nominal value (1.0 p.u.)	[78], [79], [80], [81]
14	Angle difference penalty cost (ADP)	$ADP = C_{\text{gen}}(P_G) + \beta \sum_{(i,j) \in \mathcal{E}} (\theta_i - \theta_j)^2$	Minimize generation cost plus penalty on excessive voltage angle differences across transmission lines	[82], [83]
15	Line overload penalty cost (LOP)	$LOP = \sum_g C_g(P_g) - \lambda \sum_{\ell} s_{\ell}$	Minimize generation cost with economic penalty for line thermal limit violations via slack variables s_{ℓ}	[84]

3 Constraint Extensions

3.1 Part 1

Table 6: Constraint Variants in OPF Problems (Part 1) with References

ID	Constraint Type	Formulation	Descriptions	References
1	Chance constraint	$\mathbb{P}\{h(x, \xi) \leq 0\} \geq 1 - \varepsilon$	Limit violation probability under uncertain generation/load; balance economy and risk level for high renewable penetration	[85], [86], [87], [88]
2	Robust constraint	$h(x, \xi) \leq 0, \forall \xi \in \mathcal{U}$	Maintain feasibility for all uncertainty realizations in bounded uncertainty set; provide worst-case security guarantee	[89], [90], [91], [18], [92]
3	Frequency stability constraint	$\Delta f_{\min} \geq \Delta f^{\text{req}}, \quad \text{RoCoF} \leq \text{RoCoF}^{\max}$	Limit frequency nadir and rate of change of frequency; reflect inertia and primary frequency response requirements	[93]
4	Reserve limit constraint	$-P_{R,i}^D \leq \Delta P_{G,i} \leq P_{R,i}^U, \quad -Q_{R,i}^D \leq \Delta Q_{G,i} \leq Q_{R,i}^U$	Explicitly limit dispatch adjustment capability to pre-scheduled reserve ranges for real and reactive power	[90], [59]
5	Regulation up reserve constraint	$R_j^{RU} - \sum_{i \in I_{RU} \cap Z_j} RU_i \leq 0$	Require sufficient upward regulation reserve capacity within each zone to handle load variations and forecast errors	[56], [94], [95]

3.2 Part 2

Table 7: Constraint Variants in OPF Problems (Part 2) with References

ID	Constraint Type	Formulation	Descriptions	References
6	Spinning reserve constraint	$R_j^{RU} + R_j^{SP} - \sum_{i \in I_{RU} \cap Z_j} RU_i - \sum_{i \in I_{SP} \cap Z_j} SP_i \leq 0$	Ensure adequate spinning reserve for large disturbances such as generator outages beyond regulation needs	[56], [96], [97], [98]
7	N-1 security constraint (DC)	$ B_f^{(k)} \theta^{(k)} \leq F^{\max}, \forall k \in \mathcal{C}$	Ensure no line overload after any single line outage; implement preventive security-constrained dispatch	[99], [100], [101], [102], [103]
8	N-1 security constraint (AC)	$g^{(k)}(x) = 0, \quad h^{(k)}(x) \leq 0, \quad \forall k \in \mathcal{C}$	Enforce voltage magnitude and thermal limits after any single contingency in the full AC power flow model	[104], [101], [105], [106], [107], [108]
9	Voltage stability constraint	$L_i(V, \theta) \leq L^{\max}, \quad \forall i \in N_B$	Prevent voltage instability by constraining voltage stability indices such as L-index below critical threshold	[64], [44], [43], [41], [71]

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